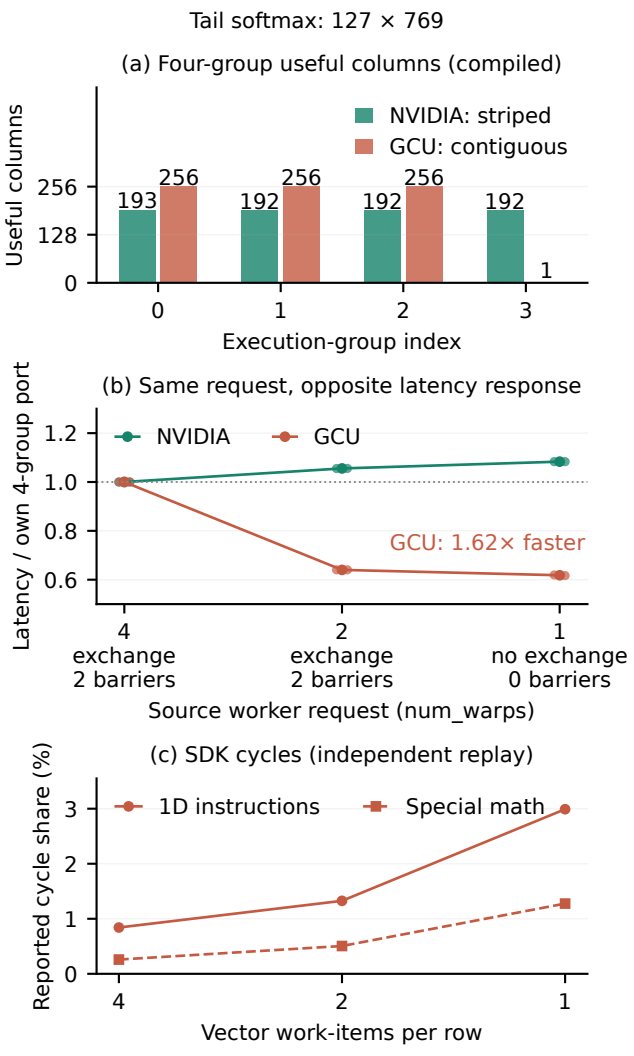
Tail softmax: 127 × 769
(a) Four-group useful columns (compiled)
Useful columns
NVIDIA: striped
GCU: contiguous
256
128
0
193
256
192
256
192
256
192
1
0
1
2
3
Execution-group index
(b) Same request, opposite latency response
Latency / own 4-group port
NVIDIA
GCU
1.2
1.0
0.8
0.6
GCU: 1.62× faster
4
exchange
2 barriers
2
exchange
2 barriers
1
no exchange
0 barriers
Source worker request (num_warps)
(c) SDK cycles (independent replay)
Reported cycle share (%)
1D instructions
Special math
3
2
1
0
4
2
1
Vector work-items per row